

JEE Main - 7 | JEE - 2023 (Gen 1 & 2)

Date: 14/12/2021

Maximum Marks: 300

Timing: 04:00 PM - 09:00 PM

Duration: 3.0 Hrs

General Instructions

1. The test is of **3 hours** duration and the maximum marks is **300**.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is an **integer** ranging from 0 to 99. (both inclusive)
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

PART - I : PHYSICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

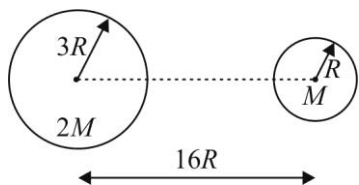
1. A planet of mass m moves along an ellipse around the sun so that maximum and minimum distances from the sun are equal to a and b respectively. The angular momentum of this planet relative to the centre of the sun will be (Mass of the Sun = M)

(A) $m\sqrt{\frac{GMa}{(a+b)}}$ (B) $m(a+b)\sqrt{\frac{GM}{ab}}$
 (C) $m\sqrt{\frac{GMab}{(a+b)}}$ (D) $mab\sqrt{\frac{GM}{(a+b)^3}}$

2. A circular platform is mounted on a vertical frictionless axle. Its radius is $r = 2\text{m}$ and its moment of inertia is $I = 200 \text{ kg-m}^2$. It is initially at rest. A 70 kg man stands on the edge of the platform and begins to walk along the edge at speed $v_0 = 10 \text{ m/s}$ relative to the ground. The angular velocity of the platform is:

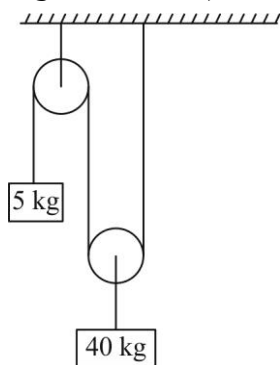
(A) 1.2 rad/sec (B) 0.4 rad/sec (C) 2.0 rad/sec (D) 0.7 rad/sec

3. Two spherical bodies of mass $2M$ and M and of radii $3R$ and R respectively are held at a distance $16R$ from each other in free space. When they are released, they start approaching each other due to the gravitational force of attraction. Then their velocities at the time of impact will be respectively:



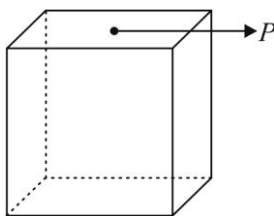
(A) $\sqrt{\frac{GM}{8R}}, 2\sqrt{\frac{GM}{8R}}$ (B) $\frac{3}{2}\sqrt{\frac{GM}{R}}, 3\sqrt{\frac{GM}{R}}$
 (C) $\sqrt{\frac{GM}{2R}}, \frac{1}{2}\sqrt{\frac{GM}{R}}$ (D) $\sqrt{\frac{GM}{3R}}, \sqrt{\frac{GM}{R}}$

4. The acceleration of centre of mass of the system shown in figure will be: (Pulleys and strings are massless, $g = -10 \text{ m/s}^2$)

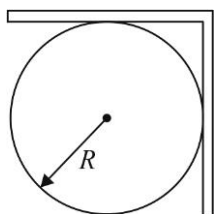


(A) -5 m/s^2 (B) $-\frac{10}{3} \text{ m/s}^2$ (C) $\frac{5}{3} \text{ m/s}^2$ (D) 10 m/s^2

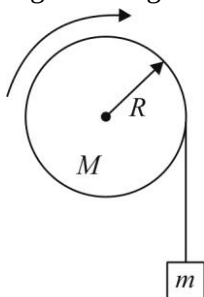
5. A solid cube is placed on a horizontal surface. The coefficient of friction between them is μ , where $\mu < \frac{1}{2}$. A variable horizontal force is applied on the cube's upper face, perpendicular to one edge and passing through its mid-point as shown in the figure. The maximum acceleration with which it can move without toppling is:



- (A) $g(1-2\mu)$ (B) $g(1+2\mu)$ (C) $g/(1-2\mu)$ (D) $g/(1+2\mu)$
6. A plate bent at right angles along its centre line, is placed onto a horizontal fixed cylinder of radius R as shown in the figure. The minimum coefficient of friction between the cylinder and the plate so that plate does not slip off the cylinder will be:



- (A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$ (C) $\sqrt{2}-1$ (D) $1-\frac{1}{\sqrt{2}}$
7. In a simple pendulum, the breaking strength of the string is double the weight of the bob. The bob is released from rest when the string is horizontal. The string breaks when it makes an angle θ with the vertical, where?
- (A) $\theta = \cos^{-1} \frac{1}{3}$ (B) $\theta = 60^\circ$ (C) $\theta = \cos^{-1} \left(\frac{2}{3} \right)$ (D) $\theta = 0^\circ$
8. Three particles, each of mass m , are located at the vertices of an equilateral triangle of side d . At what speed must they move if they all revolve under the influence of their gravitational force of attraction in a circular orbit circumscribing the triangle while still preserving the equilateral triangle?
- (A) $\sqrt{\frac{2GM}{3d}}$ (B) $\sqrt{\frac{2Gm}{d}}$ (C) $\sqrt{2Gmd}$ (D) $\sqrt{\frac{Gm}{d}}$
9. A block of mass 2 kg hangs from the rim of a wheel of radius 0.5 m . On releasing from rest the block falls through 5 m height in 2 seconds. The moment of inertia of the wheel will be:



- (A) 1.5 kg-m^2 (B) 2.1 kg-m^2 (C) 1.25 kg-m^2 (D) 0.50 kg-m^2

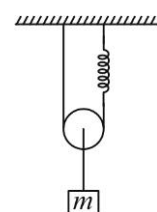
10. The system given in figure is released from rest with the spring initially stretched by x . The velocity v of the block after it has dropped a distance $\frac{x}{2}$ will be: (Pulley is massless)

(A) $\left(\frac{4mgx - 9kx^2}{4m}\right)^2$

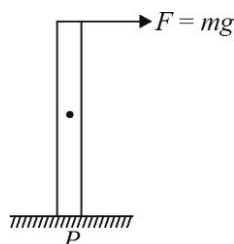
(B) $\left(\frac{mgx - 3kx^2}{m}\right)^{1/2}$

(C) $\left(\frac{mgx - kx^2}{2m}\right)^{1/2}$

(D) Zero



11. A uniform rod of mass m and length ℓ is held stationary in vertical position on a smooth horizontal surface as shown. What will be the angular acceleration of the rod in this position when a horizontal force $F = mg$ is applied at the upper end?



(A) $\frac{3g}{5\ell}$

(B) $\frac{6g}{\ell}$

(C) $\frac{7g}{5\ell}$

(D) $\frac{3g}{7\ell}$

12. Two satellites A and B revolve round a planet in coplanar circular orbits in the same sense. Their periods of revolution are 8 hour and 27 hour respectively. The radius of the orbit of A is 4×10^4 km. When B is closest to A, the angular speed of B as observed by an astronaut in A will be:

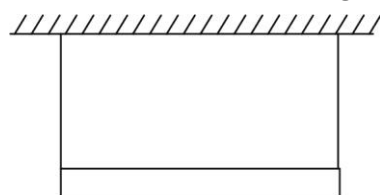
(A) $\frac{2\pi}{3}$ rad/hour

(B) $\frac{\pi}{15}$ rad/hour

(C) $\frac{\pi}{17}$ rad/hour

(D) $\frac{\pi}{8}$ rad/hour

13. A uniform rod of length ℓ and mass M is suspended on two vertical inextensible strings as shown in the figure. The tension T in the left string at the instant when right string snaps will be:



(A) $\frac{mg}{4}$

(B) $\frac{mg}{3}$

(C) $\frac{mg}{12}$

(D) mg

14. A ball of mass m moving with a speed u undergoes a head-on elastic collision with a ball of mass nm initially at rest. The fraction of initial energy transferred to the heavier ball is:

(A) $\frac{n}{1+n}$

(B) $\frac{n}{(1+n)^2}$

(C) $\frac{2n}{(1+n)^2}$

(D) $\frac{4n}{(1+n)^2}$

15. Two blocks A and B of mass m and $2m$ respectively are connected by a massless spring of force constant k . They are placed on a smooth horizontal plane. They are stretched by an amount x and then released. The relative velocity of the blocks when the spring comes to its natural length is:

(A) $x\sqrt{\frac{3k}{2m}}$

(B) $x\sqrt{\frac{2k}{3m}}$

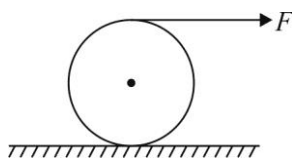
(C) $x\sqrt{\frac{2k}{m}}$

(D) $x\sqrt{\frac{3k}{m}}$

16. A plate remains in equilibrium in air when n bullets are fired per second on it. The mass of each bullet is m and it strikes the plate with speed v . The coefficient of restitution is 0.5. The mass of the plate is:

(A) $M = \frac{3mnv}{2g}$ (B) $M = \frac{mnv}{g}$ (C) $M = \frac{2mnv}{3g}$ (D) $M = \frac{2mnv}{g}$

17. A force F acts tangentially at the highest point of a solid sphere of mass m kept on a rough horizontal plane. If the sphere rolls without slipping, the acceleration of the centre of the sphere will be:



(A) $\frac{5F}{3m}$ (B) $\frac{10F}{7m}$ (C) $\frac{3F}{7m}$ (D) $\frac{7F}{2m}$

18. A smooth sphere is moving on horizontal surface with velocity vector $3\hat{i} + 3\hat{j}$ immediately before it hits a vertical wall. The wall is parallel to $\hat{i}$ vector and the coefficient of restitution between the sphere and the wall is $e = \frac{1}{3}$. The velocity vector of the sphere after it hits the wall is:

(A) $\frac{\hat{i}}{3} + \hat{j}$ (B) $-\hat{i} - \frac{\hat{j}}{3}$ (C) $3\hat{i} - \hat{j}$ (D) $-3\hat{i} - 3\hat{j}$

19. A uniform chain of length L and mass M overhangs a horizontal table with its two third part on the table. The friction coefficient between the table and the chain is μ . The work done by the friction during the period the chain slips off the table is:

(A) $-\frac{1}{4}\mu MgL$ (B) $-\frac{2}{9}\mu MgL$ (C) $-\frac{4}{9}\mu MgL$ (D) $-\frac{6}{7}\mu MgL$

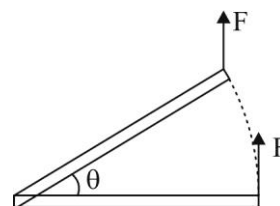
20. A thin horizontal uniform rod AB of mass m and length L can rotate freely about a vertical axis passing through its end A . At a certain moment, the end B starts experiencing a constant force F which is always perpendicular to the original position of the stationary rod and directed in a horizontal plane. The angular velocity of the rod as a function of its rotation angle θ measured relative to the initial position will be:

(A) $\omega = \sqrt{\frac{3F \sin \theta}{2mL}}$

(B) $\omega = \sqrt{\frac{6F \sin \theta}{mL}}$

(C) $\omega = \sqrt{\frac{5F \cos \theta}{mL}}$

(D) $\omega = \sqrt{\frac{F \cos \theta}{3mL}}$

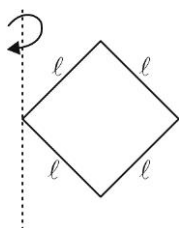


SPACE FOR ROUGH WORK

SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 99 (both inclusive).

21. Displacement of particle of mass 1 kg moving in a straight line varies with time as $S = (2t^3 + 5)m$. Impulse of the force acting on the particle over a time interval between $t = 0$ and $t = 1\text{sec}$ will be _____ (in N-s).
22. The moment of inertia of a system of four rods, each of length ℓ and mass m , about the axis shown is $\frac{N}{24}m\ell^2$. The value of N will be _____.



23. A body of mass m makes an elastic collision with another identical body at rest. Just after collision the angle between the velocity vector of one body with the initial line of motion is 15° then the angle between velocity vector of the other body with the initial line of motion will be _____ (in degrees).
24. The energy required to shift a satellite from orbital radius r to orbital radius $2r$ is E . The energy required to shift the satellite from orbital radius $2r$ to orbital radius $3r$ is $\frac{E}{n}$. The value of n will be _____.
25. A wheel rotating at an angular speed of 20 rad/sec is brought to rest by a constant torque in 4.0 seconds. If the moment of inertia of the wheel about the axis of rotation is 0.20 kg-m^2 , then work done by the torque in the first two seconds will be _____ (in Joule).

SPACE FOR ROUGH WORK

PART - II : CHEMISTRY**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- Which of the following salts undergoes anionic hydrolysis ?
(A) CuSO_4 (B) NH_4Cl (C) AlCl_3 (D) K_2CO_3
- In an aqueous solution $10^{-2} \text{ M Na}_2\text{SO}_4$ and 10^{-2} M NaI are present. Now pure $\text{Pb}(\text{NO}_3)_2$ is added gradually, then calculate concentration of SO_4^{2-} when PbI_2 start precipitating in solution $K_{\text{sp}}(\text{PbI}_2) = 10^{-9}$ and $K_{\text{sp}}(\text{PbSO}_4) = 10^{-8}$.
(A) 10^{-2} M (B) 10^{-3} M (C) 10^{-6} M (D) 10^{-5} M
- Which of the following is most soluble in water ?
(A) $\text{MnS}(K_{\text{sp}} = 8 \times 10^{-37})$ (B) $\text{ZnS}(K_{\text{sp}} = 7 \times 10^{-16})$
(C) $\text{Bi}_2\text{S}_3(K_{\text{sp}} = 5 \times 10^{-70})$ (D) $\text{Ag}_2\text{S}(K_{\text{sp}} = 6 \times 10^{-51})$
- 100 mL of 0.02 M benzoic acid ($\text{pK}_a = 4.2$) is titrated using 0.02 M NaOH. pH after 50 mL and 100 mL of NaOH have been added are :
(A) 3.50, 7 (B) 4.2, 7 (C) 4.2, 8.1 (D) 4.2, 8.25
- $\text{M}(\text{OH})_x$ has $K_{\text{sp}} = 4 \times 10^{-12}$ and solubility 10^{-4} M . The value of x is :
(A) 1 (B) 2 (C) 3 (D) 4
- How many grams of HCl should be dissolved in sufficient water to get 500 mL of an aqueous solution of pH, 2.0 ?
(A) 0.01 (B) 0.005 (C) 0.1825 (D) 0.365
- Which of the following solutions would have pH equal to pK_b of NH_3 ?
(A) 100 mL of 0.2 M HCl + 100 mL of 0.4 M NH_3
(B) 50 mL of 0.1 M HCl + 50 mL of 0.2 M NH_3
(C) 100 mL of 0.3 M HCl + 100 mL of 0.6 M NH_3
(D) None of these
- An acid-base indicator which is a weak acid has a pK_{In} value = 5.45. At what concentration ratio of sodium acetate to acetic acid would the indicator show a colour half-way between those of its acid and conjugate base forms ? [pK_a of acetic acid = 4.75, $\log 5 = 0.7$]
(A) 4 : 1 (B) 6 : 1 (C) 5 : 1 (D) 3 : 1
- The addition of sodium acetate to acetic acid solution will cause :
(A) increase in its pH value
(B) decrease in its pH value
(C) no change in pH value
(D) change in pH which cannot be predicted

10. In aqueous solution the ionization constants for carbonic acid (H_2CO_3) are :
 $K_1 = 4.2 \times 10^{-7}$ and $K_2 = 4.8 \times 10^{-11}$
 Select the correct statement for a saturated 0.034 M solution of the carbonic acid.
 (A) The concentration of CO_3^{2-} is 0.034 M
 (B) The concentration of CO_3^{2-} is greater than that of HCO_3^-
 (C) The concentration of H^+ and HCO_3^- are approximately equal
 (D) The concentration of H^+ is double that of CO_3^{2-}
11. Which of the following options is incorrect ?
 (A) pH at half equivalent point of a weak acid (HA) is equal to pK_a of weak acid
 (B) An aqueous solution of sodium acetate will be basic due to anionic hydrolysis
 (C) pH of pure water at 90°C is 7
 (D) On dilution of a solution of a weak acid, degree of ionisation will increase but $[\text{H}^+]$ will decrease
12. Which of the following relations is correct ?
 (A) $\Delta G^\circ = RT \ln K_{\text{eq}}$ (B) $[\text{H}_3\text{O}^+] = 10^{\text{pH}}$
 (C) $\log \frac{K_{w2}}{K_{w1}} = \frac{\Delta H}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$ (D) $\Delta G = -RT \ln K_{\text{eq}}$
13. At 25°C , the dissociation constants of acid HA and base BOH in aqueous solution is same. The pH of 0.01 M solution of HA is 5.0. The pH of 0.1 M solution of BOH is :
 (A) 5.0 (B) 9.0 (C) 9.5 (D) 8.5
14. 10^{-6} M HCl is diluted to 100 times. Its pH is approximately equal to :
 (A) 6 (B) 8 (C) 6.95 (D) 9.5
15. The pH at which water is maximum dissociated at 25°C , is :
 (A) 2 (B) 7 (C) 10 (D) 14
16. The solubility of $\text{Pb}(\text{OH})_2$ in water is 6.0×10^{-6} M. The solubility of $\text{Pb}(\text{OH})_2$ in a buffer solution of pH = 8 is :
 (A) 8.64 M (B) 2.16×10^{-16} M
 (C) 8.64×10^{-16} M (D) 8.64×10^{-4} M
17. Carbonic acid (H_2CO_3), a diprotic acid has $K_{a1} = 4.0 \times 10^{-7}$ and $K_{a2} = 7.0 \times 10^{-11}$. What is the $[\text{CO}_3^{2-}]$ of a 0.025 M solution of carbonic acid ?
 (A) 5.5×10^{-9} (B) 5.5×10^{-8} (C) 7.0×10^{-9} (D) 7.0×10^{-11}
18. 0.1 mole of CH_3NH_2 (pK_b is 3.4 at 25°C) is mixed with 0.08 mole HCl and diluted to 1 litre then which of the following is incorrect? ($\log 2 = 0.3$)
 (A) $[\text{H}^+] \approx 10^{-10}$ (B) pH = 10.1
 (C) pOH = 9.95 (D) It will act as a buffer solution

19. If pK_b for fluorine ion at 25°C is 10.3, the ionization constant of hydrofluoric acid in water at this temperature is : ($\log 2 = 0.3$)
- (A) 2×10^{-4} (B) 2×10^{-3} (C) 2×10^{-5} (D) 5×10^{-11}
20. A solution containing NH_4Cl and NH_3 has a hydroxide-ion concentration of 10^{-6}M . Which of the following hydroxides may be precipitated when this solution is added to an equal volume of 0.1 M solution of a metal ion ?
- (A) $\text{Ba}(\text{OH})_2 (K_{sp} = 10 \times 10^{-4})$ (B) $\text{Ca}(\text{OH})_2 (K_{sp} = 2.2 \times 10^{-5})$
- (C) $\text{Mg}(\text{OH})_2 (K_{sp} = 3.0 \times 10^{-12})$ (D) $\text{Fe}(\text{OH})_2 (K_{sp} = 8.0 \times 10^{-16})$

SPACE FOR ROUGH WORK

SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 99 (both inclusive).

21. Calculate solubility of $\text{Ca}_3(\text{PO}_4)_2$ ($K_{sp} = 10^{-15}$) in presence of 0.1 M CaCl_2 solution.
- If your answer is $x \times 10^{-y}$ where x is single digit then fill y-x in OMR sheet, e.g., if answer is 2×10^{-3} then answer is 1.
22. x (mL) of water is added to 300 mL of a 0.2 M solution of CH_3COOH for the degree of dissociation of the acid to double? Value of $\frac{x}{100}$ is? [K_a of acetic acid 10^{-5}]
23. If a solution is 0.1M with respect to H_2S and 0.5M with respect to HCl then the concentration of $[\text{S}^{2-}]$ in the solution is $x \times 10^{-22}\text{M}$. The value of x is : Given : $\left. \begin{array}{l} K_{a1} = 10^{-7} \\ K_{a2} = 10^{-14} \end{array} \right\} \text{H}_2\text{S}$
24. The pH range of a basic indicator (InOH) is 3.4–4.6. Determine the ratio $[\text{In}^+]/[\text{InOH}]$ above which the solution appears only in the colour of In^+ ? ($\log 2 = 0.3$)
25. The ionic product of heavy water (D_2O) is 1.0×10^{-16} at 7°C . The P^D value of heavy water at 7°C is_____.

SPACE FOR ROUGH WORK

PART - III : MATHEMATICS

100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- If $x - \frac{1}{x} = \sqrt{2}i$ then value of $\left(x^{2019} - \frac{1}{x^{2019}}\right)$ is equal to:
 (A) $\sqrt{2}i$ (B) $-\sqrt{2}i$ (C) $\frac{1}{\sqrt{2}i}$ (D) $\frac{-1}{\sqrt{2}i}$
- The number of positive integral values of 'a' for which the equation $(1+a^2)x^2 - 2(1+3a)x + (1+8a) = 0$ has real roots, is:
 (A) 3 (B) 2 (C) 1 (D) 0
- Domain of the function $f(x) = \frac{\sqrt{1-x^2}}{|x|} + \log(x^2 - 4)$ is:
 (A) ϕ (B) $\mathbb{R}$ (C) $[-1, 1] - \{0\}$ (D) $[-1, 2] - \{0\}$
- If points A, B and C are represented by z_1, z_2 and z_3 on argand plane and all of them lie on $|z| = 6$, then orthocentre of $\triangle ABC$ is given by:
 (A) O (B) $\frac{z_1 + z_2 + z_3}{3}$
 (C) $z_1 + z_2 + z_3$ (D) information is insufficient
- If $f(x) = \cos\left(x - \frac{\pi}{4}\right) - \cos\left(x + \frac{\pi}{4}\right) + \sin^2 x$, then range of $f(x)$ is:
 (A) $\left[-\frac{1}{2}, \sqrt{2} - 1\right]$ (B) $[0, \sqrt{2} + 1]$ (C) $[0, 1]$ (D) $\left[\frac{-1}{2}, \sqrt{2} + 1\right]$
- If ' α ' ($\alpha \neq 1$) is complex cube root of unity, then $(1+\alpha)^2 + (1+\alpha^2)^2 + (1+\alpha^4)^2 + (1+\alpha^6)^2$ is equal to:
 (A) 0 (B) $\alpha + 3$ (C) -2 (D) $3 + \alpha^2$
- $[x]^2 + [y]^2 = 6[x] + 4[y] + 13$ where $[k]$ represents greatest integer less than or equal to k . The value of $[x + y]$ is equal to ' α ', then:
 (A) $\alpha = 5$ only (B) $\alpha \in \{5, 6\}$ (C) $\alpha \in \{6, 7\}$ (D) $\alpha \in \{5, 6, 7\}$
- How many words can be made using all the letters v, i, d, y, a, b, e, s, t only once so that neither 'vidya' nor 'best' patterns appear anywhere in the word formed?
 (A) $9! - 5! - 6! - 2$ (B) $9! - 5! - 6!$ (C) $9! - 5! - 6! + 1$ (D) $9! - 5! - 6! + 2$
- If $\left({}^{10}C_r\right) + 2\left({}^{10}C_{r+1}\right) = \left({}^{10}C_{r-1}\right)$, then rC_3 is equal to:
 (A) 56 (B) 20 (C) 84 (D) 35
- If α and β are distinct roots of $x^2 - kx + 1 = 0$ then $(\alpha^5 + \beta^5)$ is equal to:
 (A) $k^5 - 5k^3 - 5k$ (B) $k^5 - 5k^3 + 5k$ (C) $k^5 + 5k^3 - 5k$ (D) $k^5 + 5k^3 + 5k$

11. Which of the following function is an odd function?
- (A) $(f(x) + f(-x))(g(x) - g(-x))^2$ (B) $\frac{f(x) - f(-x)}{g(x) - g(-x)}$
- (C) $\log\left(\frac{2+x}{2-x}\right)$ (D) $\left(\frac{e^x + 1}{e^x - 1}\right)\sin x$
12. If $\alpha = C_0 + 6C_1 + 26C_2 + 126C_3 + 626C_4 + \dots$ till n terms where $C_r = {}^nC_r$, then α is equal to:
- (A) $6^n + 2^n - 1$ (B) $6^n + 2^n$ (C) $6^n - 2^n + 1$ (D) $6^n - 2^n - 1$
13. A basket contains 4 oranges, 5 apples and 6 mangoes. In how many ways can a person make a selection of fruits, if at least one fruit have to be selected and fruits of same type are considered to be identical.
- (A) 210 (B) 209 (C) 120 (D) 119
14. If exactly two values of x satisfy $|x^2 - 2| = a$ then:
- (A) $a < 0$ (B) $a \in (0, 2)$ (C) $a \geq 2$ (D) $a \in (2, \infty) \cup \{0\}$
15. The term independent of ' x ' in the expansion $(3x + 2)\left(2x - \frac{1}{x}\right)^{17}$ is equal to:
- (A) $3({}^{17}C_9)(2^8)$ (B) $-3({}^{17}C_9)(2^8)$ (C) $-3({}^{17}C_8)(2^9)$ (D) $3({}^{17}C_9)(2^9)$
16. If both distinct and real roots of the equation $x^2 - ax + (5a - 1) = 0$ are greater than 8, then:
- (A) $a \in (16, 21)$ (B) $a \in (16, 10 + 4\sqrt{6})$
- (C) $a \in (10 + 4\sqrt{6}, 21)$ (D) $a \in (21, \infty)$
17. If $f(x)$ and $g(x)$ are polynomial functions defined for $x \in \mathbb{R}$, such that $g(x) = x^2 - 5x + 4$ and $g(f(x)) = 9(x^2 + 3x + 2)$, then $f(1)$ is equal to:
- (A) 10 or -5 (B) 10 or 5 (C) -10 or -5 (D) -10 or 5
18. The remainder when $(13)^{43} + (43)^{13}$ is divided by 7 is equal to:
- (A) 1 (B) 2 (C) 0 (D) 6
19. How many words can be made using all letters of word 'BAHUBALI' if all the words start and end with vowels?
- (A) 2160 (B) 1080 (C) 1560 (D) 1800
20. If $(1 + 3x + 5x^2)^{15} = a_0 + a_1x + a_2x^2 + \dots + a_{30}x^{30}$, then $(a_0 + a_2 + a_4 + \dots + a_{30})$ is equal to:
- (A) $\frac{3^{30}}{2}$ (B) $3^{15} + 3^{30}$ (C) $\frac{3^{15}(3^{15} + 1)}{2}$ (D) $\frac{9^{15} - 3^{15}}{2}$

SPACE FOR ROUGH WORK

SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 99 (both inclusive).

1. $\operatorname{Im}\left[(3+2\sqrt{-54})^{1/2} - (3-2\sqrt{-54})^{1/2}\right]$ can be equal to $\pm 2\sqrt{a}$, $a \neq 0$ then 'a' is equal to _____.
2. The sum of the roots of the equation $2^{33x-2} + 2^{11x+2} = 2^{22x+1} + 1$ is $\frac{a}{b}$, where a and b are co-prime positive integers, then $(a+b)$ is equal to _____.
3. If $P(x) = (x^2 - 2x + 2)^9 (x^2 + 2x + 2)^{10} (x^4 + 4)$, then coefficient of x^{36} in $P(x)$ is equal to _____.
4. The number of 4 letter words (with or without meaning) that can be formed from the letters of the word 'PRACTICAL' is 'abcd' (a four digit number), then $a + b + c + d$ is equal to _____.
5. Period of the function $f(x) = \sin^2 \frac{\pi x}{5} + \cos^2 \frac{\pi x}{4}$ is equal to _____.

SPACE FOR ROUGH WORK